Semester II
Skill Based Course

Department of Economics
Delhi School of Economics

Report
Forecasting Norway's GDP per Capita
A Time Series Approach

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I.Introduction

Norway is a wealthy country in Northern Europe consistently ranked among the top nations in the world by GDP per capita. Its economy is driven mainly by oil and natural gas production which expanded significantly after the discovery of North Sea oil in the early 1970s. Apart from petroleum other important sectors include aquaculture, maritime shipping and financial services. The government has managed its oil revenues through a large sovereign wealth fund known as the Government Pension Fund Global which has helped shield the economy from external shocks and maintain long run stability.

Over the past several decades Norway's GDP per capita has shown consistent upward trend. However the series has not been without disruptions. Sharp falls in global oil prices particularly in the mid 1980s and around 2014 to 2016 caused noticeable slowdowns in growth. The 2008 Global Financial Crisis led to a mild contraction compared to most other advanced economies while the COVID-19 pandemic of 2020 caused a short but sharp decline before a quick recovery.

This project examines the trends in Norway's GDP per capita using data sourced from the World Bank covering the period 1960 to 2024. The dataset provides a long run annual series suitable for forecasting and consists of two main components the year and the corresponding GDP per capita value expressed in constant US dollars. GDP per capita is obtained by dividing a country's total output by its population and is widely used as a measure of economic health, productivity and living standards. A visual inspection of the series reveals a strong upward trend with fluctuations arising from the economic shocks described above and negligible seasonality given the annual frequency of the data.

II.Data Visualisation & Description

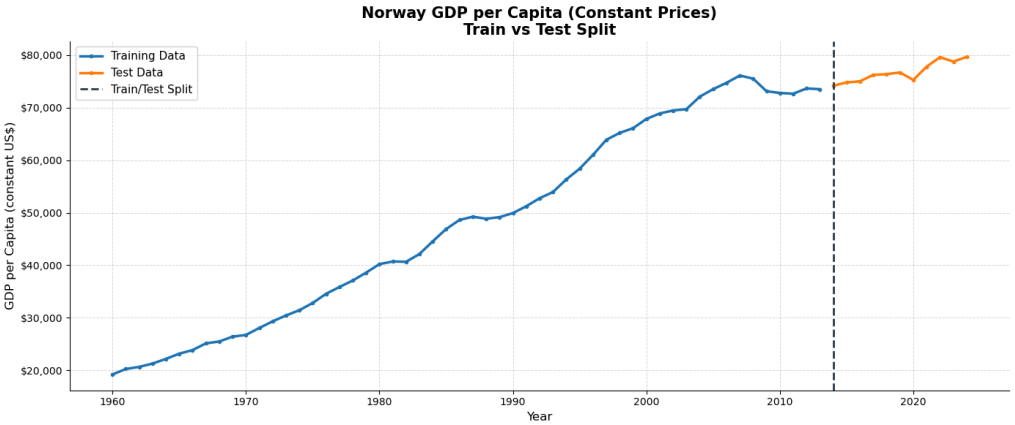


Figure 1: Time series of the Norway GDP per capita from 1960 to 2024

- **Trend**
  From roughly $20,000 in 1960 to close to $80,000 by 2024 the series reflects a near fourfold increase driven primarily by the transformation of Norway's economy following the discovery of North Sea oil in the early 1970s. What is notable is not just the magnitude of growth but its consistency. Unlike many resource dependent economies Norway avoided the boom and bust pattern typically associated with commodity dependence largely due to the buffering role played by the Government Pension Fund Global. The trend therefore has a clear structural explanation.

- **Volatility and Shocks**
  The series is not smooth throughout. Three events introduced visible disruptions: oil price collapse of the mid 1980s, the Global Financial Crisis of 2008 to 2009 and the COVID-19 contraction of 2020. In each case the deviation from trend was relatively short lived and the series returned to its upward path. This pattern of sharp but temporary shocks followed by recovery is an important feature as it means forecasting models need to track the trend well rather than overfit to short run fluctuations.

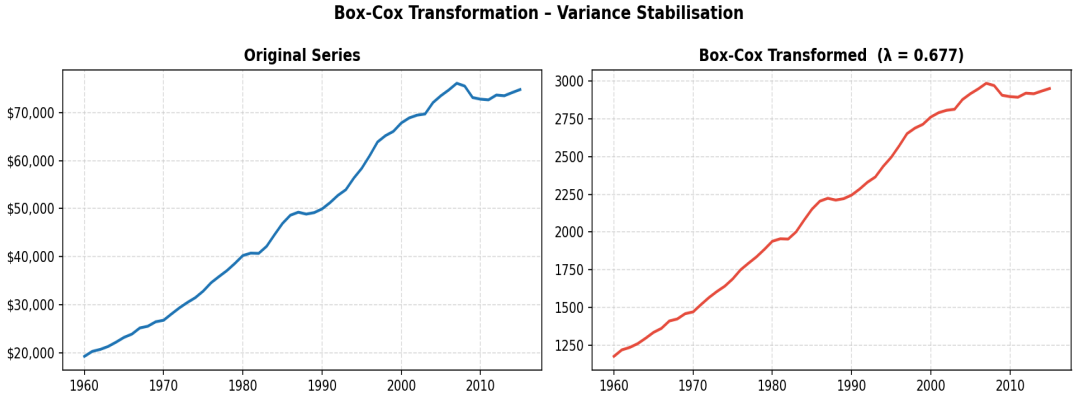
- **Stationarity**
  The series is clearly non stationary in levels. The mean rises over time and variance increases alongside it. This has direct implications for model selection as methods that assume a constant mean such as the historical average are fundamentally unsuited to this data. Models that either extrapolate the trend directly or difference the series to achieve stationarity are the appropriate tools here. For using the ARIMA forecasting method we need the series to be stationary.

III.Methodology

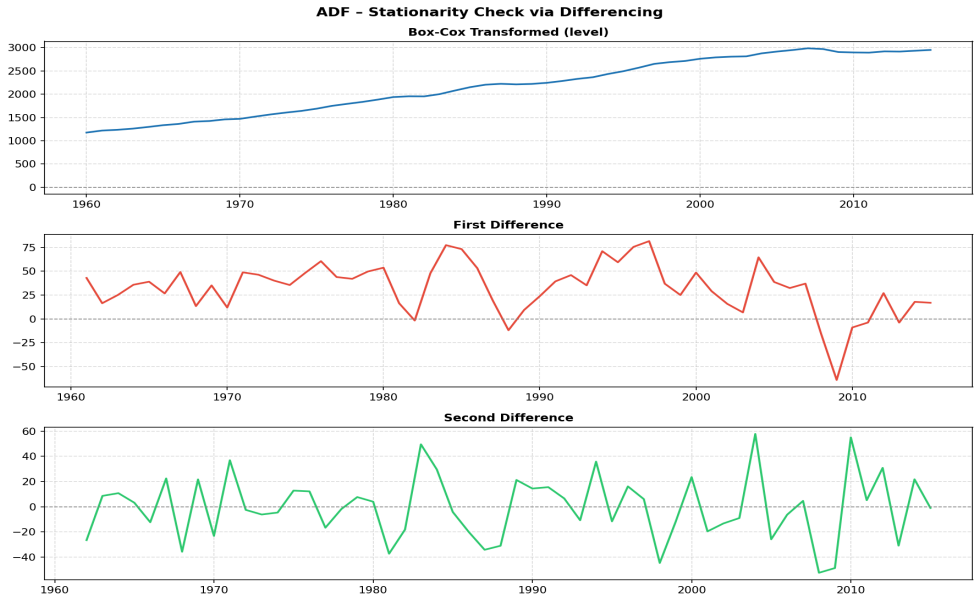
The study intends to perform a time series forecasting to predict trends in the GDP per capita of the country of Norway. The approach to this purpose is structured into major steps which are data preparation, transformation of series, model construction, forecasting and evaluation.

- **Data Preparation**
  The data consists of time series observations of GDP per capita for the period 1960 to 2024 sourced from the world bank's data records. The data went through a preliminary cleaning and structuring process.
  In an effort to ensure robustness of the model the years from 1960 to 2015 were used as the training set which were used in estimating the model's parameters and the subsequent years from 2016 to 2024 were used as the testing data set for the purpose of validation.
- **Transformation of Series for ARIMA Model**

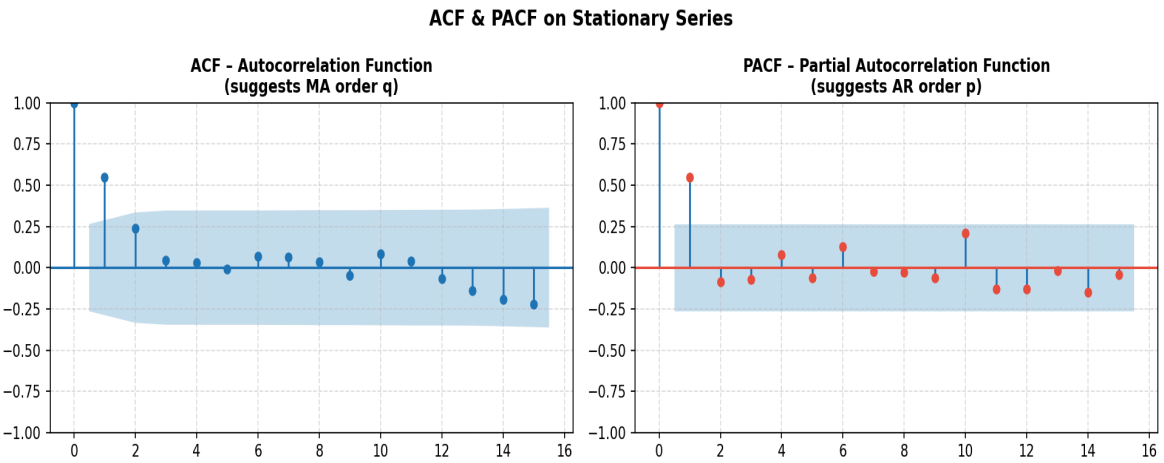
➔ **Box-Cox Transformation to stabilize the variance.**



➔ **First Differencing to make the mean constant.**



➔ **ACF/PACF Plot to check for stationarity**



- **Forecasting Models**

The following models were employed in the analysis in order to facilitate predictions which are widely used in forecasting applications:

➔ **Mean Method:** The basic assumption underlying this model is that future values will equal the historical average of the observations used in the training dataset.

- ➔ **Naive Method:** This model assumes that the value of the variable in question for the subsequent periods will equal the latest observed value.
- ➔ **Drift Method:** This model can be considered an extension of the naive method. The underlying assumption is that the subsequent values in the series will follow a linear trend. The linear trend line is incorporated based on the recently observed changes. The future values are calculated by extrapolating a trend graph using the past data.
- ➔ **Exponential Smoothing Method (ETS):** This accounts for trend components and any seasonality effect present in the data. It uses the weighted average of the past data. The model is particularly useful when the data displays systematic patterns periodically. It is used to capture both the level and trend effect of the time series data. The model assigns greater weight to recent observations while progressively reducing the influence of older data.
- ➔ **Autoregressive Integrated Moving Averages (ARIMA) Model:** This is a time series model that explains current GDP using past values and past errors. It works on the changes in the data and removes trends to make the series stable allowing it to capture underlying patterns effectively.

In order to check the stationarity of the data series a Kwiatkowski Phillips Schmidt Shin testKPSS (KPSS) test was employed for the original and transformed data.

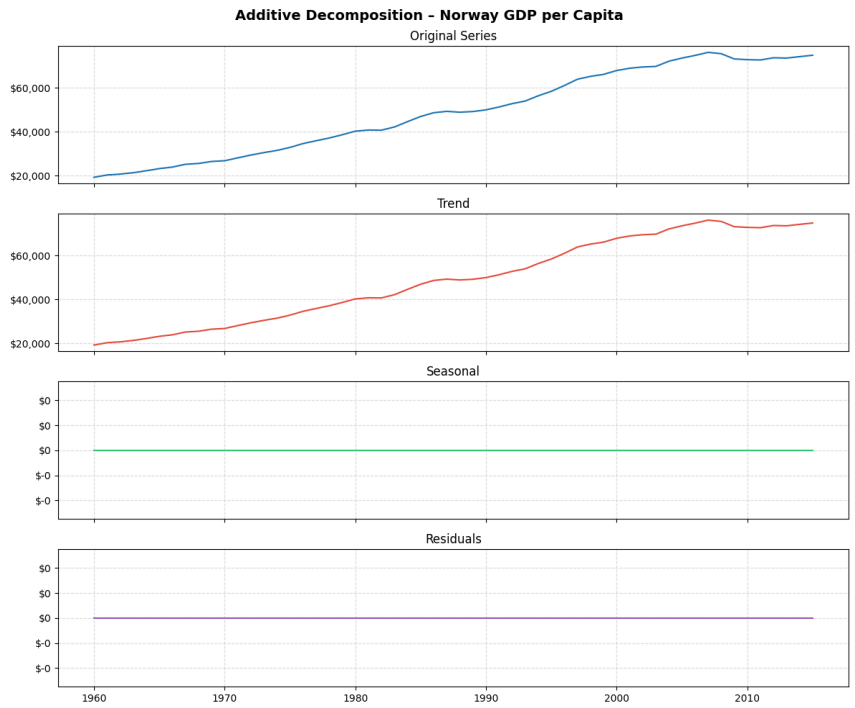
KPSS Statistic	1.166991518	0.01
P-value	0.3020474029	0.1

- **Model Evaluation**

  The accuracy of forecasts so generated were evaluated based on their performance in terms of the following 3 metrics:
  - ➔ **Mean Absolute Error (MAE):** This metric measures the average magnitude of forecast errors i.e. average absolute difference between predicted and actual values.
  - ➔ **Root Mean Squared Error (RMSE):** This metric puts more weightage on larger errors and penalises the smaller ones lesser. It accomplishes this by squaring the errors. Measures the square root of the average squared forecast errors.
  - ➔ **Mean Absolute Percentage Error (MAPE):** This metric expresses forecast errors as percentage of actual values. This enables comparison of the errors across different scales thereby making it a measure of relative accuracy.

IV.Forecasting Evaluation

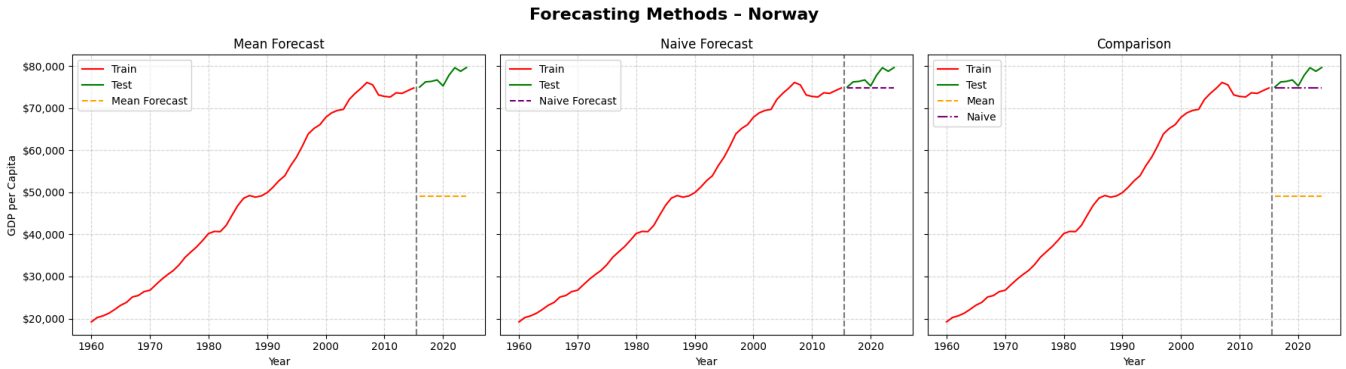
Using the data set a time series graph was plotted and decomposed into trends, seasonal and residuals as shown below



The original time series plot shows a strong upward movement. This is indicative of Norway's sustained economic growth. It is observed that the trend line follows a similar pattern with negligible seasonal and residual effects. The graphs of seasonal and residuals do not show much fluctuations. So Norway's per capita GDP is less affected by seasonal and unexplained variations. These results imply that the series is non stationary and primarily trend driven. Therefore forecasting models incorporating trend dynamics such as drift or exponential smoothing are more appropriate than models assuming a constant mean.

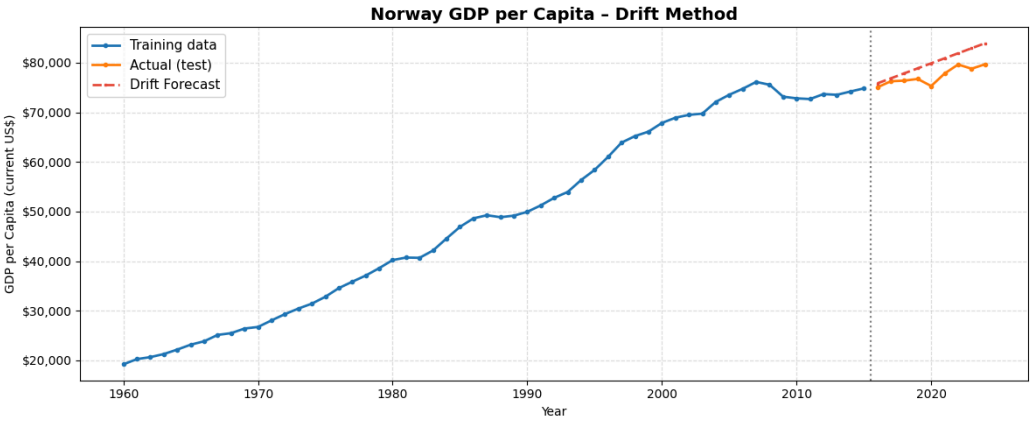
- **Historical Average and Naive Methods**

  The forecasts from historical average and naive are compared with actual values in figure given below:



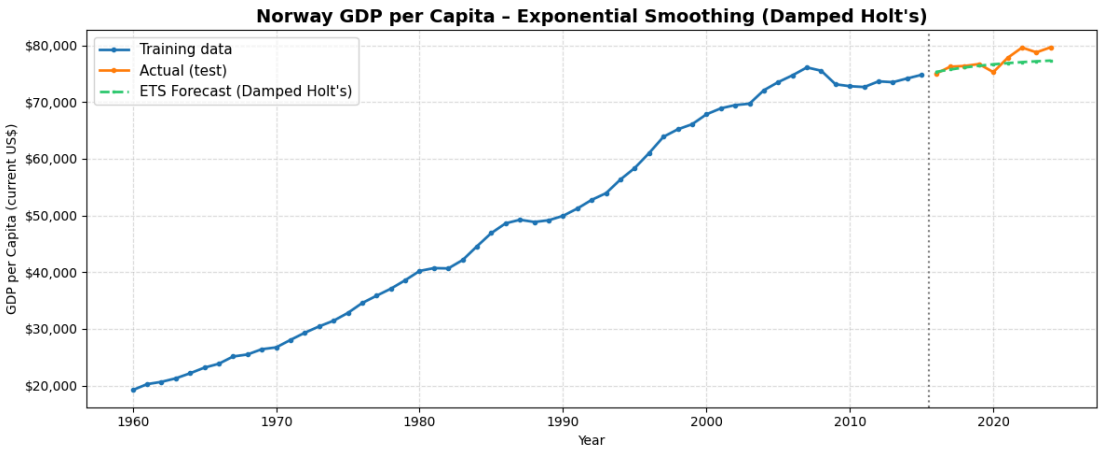
The mean method forecast is much lower than the actual values which shows that historical averages are not suitable for data where there are fluctuations. The naive method does better than historical averages in this case by predicting closer to the actual values. The results demonstrate that for non stationary series with a pronounced trend, methods that retain recent information such as the naive approach provide a more reliable baseline than those based on long run averages.

- **Drift Method**
  The results of drift method are as follows:



The drift method captures the upward trend showcased by the training data. As can be seen the forecasted values closely follow the actual data. This method does better than the mean and naive method as it incorporates both level and growth of the series. Though the method may not be suitable for data with small fluctuations but it fits well here with the strong upward movement showcased by the data.

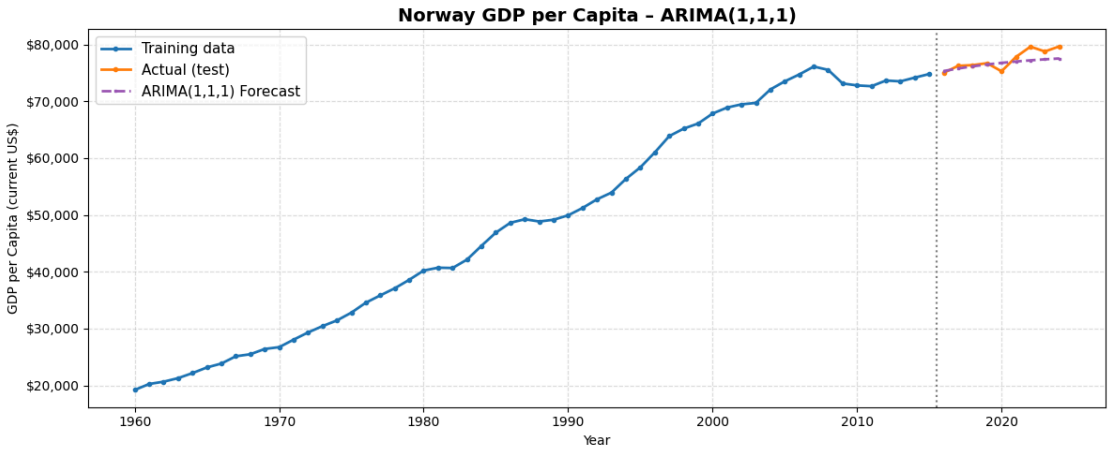
- **Exponential Smoothing**
  The results of the exponential smoothing are given below:



The forecast above is generated using damped Holt's method. It applies a damping factor to prevent the trend from growing indefinitely while capturing the level and trend of the series. ETS model produces a gradually increasing forecast which is similar to the drift method but with a smoother trajectory. The forecast aligns well with the actual data but still does not do much to capture small fluctuations.

- **ARIMA**
  For ARIMA forecasting firstly data was transformed using Box-Cox to stabilize variance. The estimated lambda value was approx 0.19 which indicates that a transformation close to the logarithmic form was appropriate. This transformation reduces heteroscedasticity and makes the series more suitable for modeling. The series still showcased a strong upward trend so differencing was applied. Based on the

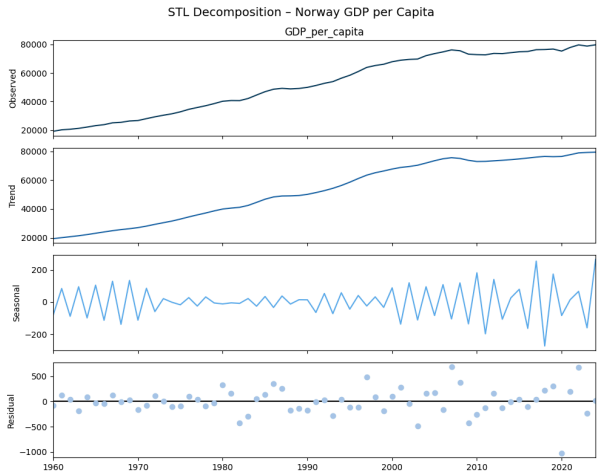
observations an ARIMA model with d = 1 and small values of p as well as q was considered appropriate for modeling the series. The forecast obtained is presented in the figure below:

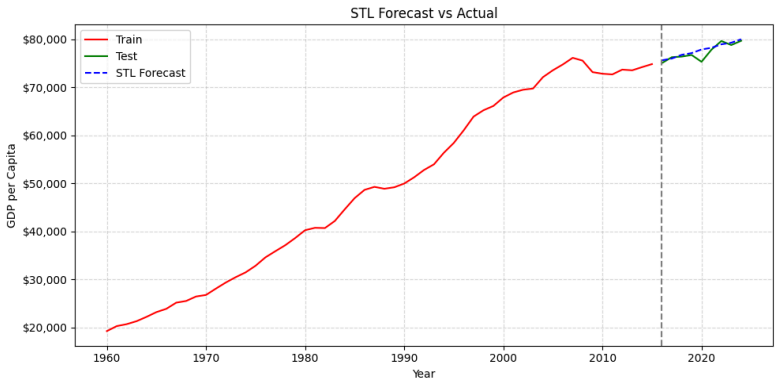


The Ljung-Box test was conducted to examine whether residuals from ARIMA have autocorrelation. With a p value > 0.05 failure to reject the null hypothesis that there is no autocorrelation in residuals. Thus the residuals behave like white noise confirming that ARIMA has captured the underlying structure of the data.

- **STL Decomposition**

  The forecast obtained from STL Decomposition is presented in the figure below:



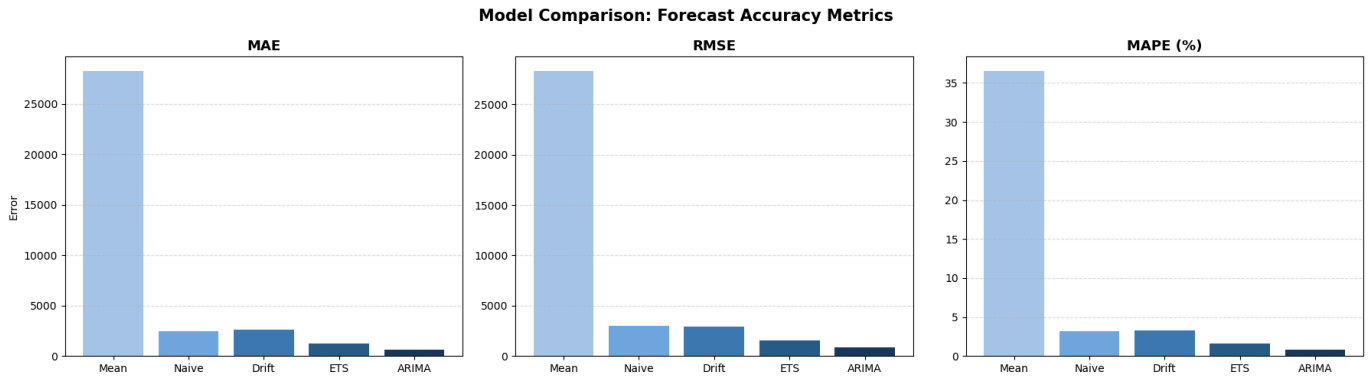


The STL forecast may look good but with careful analysis it's observed that it attempts to capture seasonal fluctuations that are almost negligent given the annual nature of data. This leads to underestimation in time periods when GDP per capita was high. Thus it fails to reflect the strong upward trend observed in the data.

V.Model Evaluation

S. No.	Model	MAE	RMSE	MAPE
1	Mean	28259.52	28308.63	36.54
2	Naive	2469.39	2979.17	3.15
3	Drift	2585.21	2947.13	3.33
4	ETS	1255.97	1565.44	1.6
5	ARIMA	610.22	896.2	0.8

The ARIMA model emerges as the best performing model among all as it has lowest error metrics. Thus chosen for the purpose of final forecast. The Mean Forecast performs the worst with extremely high error metrics which is expected as the method ignores both trend and temporal dynamics. The Naive and Drift methods perform significantly better. But both remain simplistic and fail to capture underlying dynamics. The ETS model accounts for both level and trend components so it produces substantially lower forecast errors.



VI. Economical and Statistical Significance

The mean method performed the worst with a MAPE of 36.54% because it assumes a constant average which is unsuitable for a steadily rising series like Norway's GDP per capita. Since the economy has grown significantly over time this method consistently underestimates recent values.

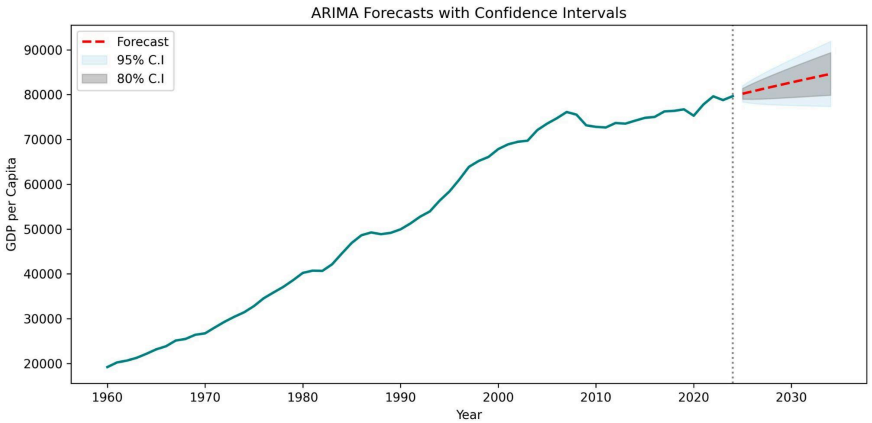
The naive and drift methods performed better with the naive method relying on the last observed value and the drift method capturing the overall upward trend. ETS improved further with a MAPE of 1.6% by giving more weight to recent data though its forecasts tend to flatten over time.

ARIMA performed the best with the lowest error of 0.8% MAPE. After transforming the data to make it stable ARIMA was able to capture the underlying pattern effectively and diagnostic tests confirmed a good model fit. This aligns with the steady and predictable growth pattern of Norway's economy.

VII. Final Forecast

Based on the comparison of all models ARIMA was found to be the best performing model on the test set recording the lowest forecast errors as the Ljung-Box test returned "white noise" for Drift and ETS as well. The Ljung-Box statistic for the ARIMA model was 0.181899. Therefore there is no autocorrelation in residuals. Hence the ARIMA Model is used to generate the forecast.

Year	Forecast (USD)
2025	80193.62942
2026	80712.58643
2027	81224.49435
2028	81729.44801
2029	82227.54102
2030	82718.86574
2031	83203.51333
2032	83681.57375
2033	84153.13578
2034	84618.28704



Using the ARIMA model Norway's GDP per capita is forecast to increase from $80,194 in 2025 to $84,618 by 2034 indicating a total rise of approximately $4,400 over the decade. The growth is gradual and steady which averages to roughly $490 per year which is consistent with Norway's historically stable long run economic path.

VIII. Conclusion

The analysis of Norway's GDP per capita from 1960 to 2024 shows a clear and consistent increase over time with only brief slowdowns during major global events such as the financial crisis and the pandemic. The data mainly follows a strong upward trend and does not show much seasonal variation. Because of this simple methods like the mean method do not work well as they ignore the changing pattern in the data. Models that take into account recent values and the overall trend provide much better results.

Year	Actual GDP	Mean	Naive	Drift	ETS	ARIMA
2016	75017.49	49019.83	74809.97	75820.88	75313.69	75397.03
2017	76247.41	49019.83	74809.97	76831.8	75716.67	75975.59
2018	76373.65	49019.83	74809.97	77842.72	76039.05	76545.76
2019	76712.29	49019.83	74809.97	78853.64	76296.96	77107.67
2020	75287.21	49019.83	74809.97	79864.56	76503.29	77661.44
2021	77812.71	49019.83	74809.97	80875.48	76668.35	78207.19
2022	79620.16	49019.83	74809.97	81886.4	76800.4	78745.02
2023	78775.75	49019.83	74809.97	82897.32	76906.04	79275.07
2024	79667.53	49019.83	74809.97	83908.24	76990.56	79797.43

Among all the models used ARIMA performs the best as it gives the lowest forecast errors and captures the pattern in the data effectively. The model also passes diagnostic checks which shows that it fits the data well. The final forecast suggests that Norway's GDP per capita will continue to grow steadily over the next ten years which is in line with its past performance. Overall the study shows that choosing the right model based on the nature of the data is important and ARIMA is a suitable method for forecasting the type of economic series obtained in the case of Norway.